

NEET (UG) 2024

National Eligibility cum Entrance Test (Undergraduate)

Date: **05 May 2024** | Test Booklet Code: **T3**

Duration: 3 hours 20 minutes | 200 MCQs | Maximum Marks: 720

General Instructions:

1. The test contains **200** multiple-choice questions from **Physics, Chemistry, Botany and Zoology** (50 questions per subject). Each subject is divided into Section A (35 compulsory) and Section B (attempt any 10 of 15).
2. Each correct response: **+4 marks**. Each incorrect response: **–1 mark**. Maximum marks: **720**.
3. In Section B, candidates need to attempt any 10 of 15 questions. If more than ten are answered, the first ten attempted will be evaluated.
4. Use of an electronic/manual calculator is prohibited.
5. The official answer key is provided at the end of this booklet.

Disclaimer: This document is a faithful reproduction of the question paper for educational and revision purposes. For the official paper and answer key, please refer to NTA / NMC.

PHYSICS**Section A — Questions 1–35 (All compulsory)**

Q1. A tightly wound 100-turn coil of radius 10 cm carries a current of 7 A. The magnitude of the magnetic field at the centre of the coil is ($\mu_0 = 4\pi \times 10^{-7}$ SI units):

- (1) 4.4 mT
- (2) 44 T
- (3) 44 mT
- (4) 4.4 T

Q2. Match List-I (Material) with List-II (Susceptibility χ): A. Diamagnetic B. Ferromagnetic C. Paramagnetic D. Non-magnetic $\leftrightarrow$ I. $\chi = 0$ II. $0 > \chi \geq -1$ III. $\chi \gg 1$ IV. $0 < \chi < \infty$ (small +ve)

- (1) A-III, B-II, C-I, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-II, B-III, C-IV, D-I
- (4) C-II, B-I, C-III, D-IV

Q3. A thermodynamic system is taken through cycle abcda (a $\rightarrow$ b at 100 kPa, b $\rightarrow$ c at 400 cm³, c $\rightarrow$ d at 300 kPa, d $\rightarrow$ a at 100 cm³). Work done by the gas along path bc is:

- (1) -90 J
- (2) -60 J
- (3) zero
- (4) 30 J

Q4. An unpolarised light beam strikes a glass surface at Brewster's angle. Then:

- (1) Both reflected and refracted light will be completely polarised
- (2) Reflected light will be completely polarised but refracted will be partially polarised
- (3) Reflected light will be partially polarised
- (4) Refracted light will be completely polarised

Q5. In an ideal transformer $N_p/N_s = 1/2$. The ratio $V_s : V_p$ is:

- (1) 1 : 1
- (2) 1 : 4
- (3) 1 : 2
- (4) 2 : 1

Q6. A logic circuit gives Y per truth table: A=0,B=0 $\rightarrow$ Y=1; A=0,B=1 $\rightarrow$ Y=0; A=1,B=0 $\rightarrow$ Y=1; A=1,B=1 $\rightarrow$ Y=0. Y = ?

- (1) B
- (2) B
- (3) $A \cdot B + \bar{A} \cdot \bar{B}$
- (4) $A \cdot B + \bar{A} \cdot \bar{B}$

Q7. In a vernier callipers, (N+1) divisions of vernier scale coincide with N divisions of main scale. If 1 MSD = 0.1 mm, the vernier constant (in cm) is:

- (1) 100 N
- (2) 10(N+1)
- (3) 1/(10N)
- (4) 1/[100(N+1)]

Q8. Maximum elongation of a 1 m steel wire if elastic limit = $8 \times 10^{11} \text{ N m}^{-2}$ and Young's modulus = $2 \times 10^{11} \text{ N m}^{-2}$:

- (1) 40 mm
- (2) 8 mm
- (3) 4 mm
- (4) 0.4 mm

Q9. A horizontal 10 N force is applied to block A (2 kg) which pushes block B (3 kg) on frictionless surface. The force exerted by A on B is:

- (1) 6 N
- (2) 10 N
- (3) zero
- (4) 4 N

Q10. If the monochromatic source in Young's double slit experiment is replaced by white light, then:

- (1) Central bright white fringe surrounded by a few coloured fringes
- (2) All bright fringes will be of equal width
- (3) Interference pattern will disappear
- (4) Central dark fringe surrounded by a few coloured fringes

Q11. Variation of $(1/\lambda^2)$ versus kinetic energy E (λ = de Broglie wavelength of a free particle) is best represented by:

- (1) Concave-up curve through origin
- (2) Straight line through origin
- (3) Concave-up sharply increasing curve
- (4) Curve increasing slowly then steeply

Q12. In the circuit (4 capacitors of $2 \mu\text{F}$ each forming a Wheatstone-like bridge), the equivalent capacitance between A and B is:

- (1) $0.5 \mu\text{F}$
- (2) $4 \mu\text{F}$
- (3) $2 \mu\text{F}$
- (4) $1 \mu\text{F}$

Q13. A bar magnet is moved towards solenoid-2 from solenoid-1. Direction of induced current in solenoid-1 and solenoid-2 respectively are:

- (1) AB and CD
- (2) BA and DC
- (3) AB and DC
- (4) BA and CD

Q14. Statements: A. For a solar cell, I-V characteristics lie in IV quadrant. B. In a reverse-biased pn junction, the (μA) current is due to majority charge carriers.

- (1) Both A and B are correct
- (2) Both A and B are incorrect
- (3) A correct, B incorrect
- (4) A incorrect, B correct

Q15. A light ray enters a right-angled prism at P with incidence 30° , travels parallel to base BC, emerges along face AC. The refractive index is:

- (1) $\sqrt{3}/4$
- (2) $\sqrt{3}/2$
- (3) $\sqrt{5}/4$
- (4) $\sqrt{5}/2$

Q16. A: Potential at axial point 2 m from a dipole of moment $4 \times 10^{-28} \text{ C}\cdot\text{m}$ is $\pm 9 \times 10^3 \text{ V}$. R: $V = \pm 2P/(4\pi\epsilon_0 r^2)$ for axial point.

- (1) A true, R false
- (2) A false, R true
- (3) Both A and R true; R is correct explanation
- (4) Both A and R true; R is NOT correct explanation

Q17. Moment of inertia of a thin rod about an axis through midpoint perpendicular to rod = $2400 \text{ g}\cdot\text{cm}^2$. Length of the 400 g rod is nearly:

- (1) 20.7 cm
- (2) 72.0 cm
- (3) 8.5 cm
- (4) 17.5 cm

Q18. Terminal voltage of a battery (emf 10 V, internal resistance 1Ω) connected to external 4Ω is:

- (1) 8 V
- (2) 10 V
- (3) 4 V
- (4) 6 V

Q19. Match Hydrogen spectral transitions with wavelengths (nm): A. $n=3 \rightarrow 2$ B. $n=4 \rightarrow 2$ C. $n=5 \rightarrow 2$ D. $n=6 \rightarrow 2$ ↔ I. 410.2 II. 434.1 III. 656.3 IV. 486.1

- (1) A-IV, B-III, C-I, D-II
- (2) A-I, B-II, C-III, D-IV
- (3) A-II, B-I, C-IV, D-III
- (4) A-III, B-IV, C-II, D-I

Q20. Correct statements about photons: A. $E = h\nu$ B. velocity = c C. $p = h\nu/c$ D. Energy and momentum are conserved in photon-electron collision E. Photon has +ve charge.

- (1) A, C and D only
- (2) A, B, D and E only
- (3) A and B only
- (4) A, B, C and D only

Q21. ${}^{288}_{82}\text{X} \rightarrow \alpha \rightarrow \text{Y} \rightarrow \text{e}^- \rightarrow \text{Z} \rightarrow \beta^- \rightarrow \text{P} \rightarrow \text{e}^- \rightarrow \text{Q}$. Mass and atomic numbers of Q are:

- (1) 288, 82
- (2) 286, 81
- (3) 280, 81
- (4) 286, 80

Q22. Displacement of a particle is $x = 2t - 1$ (SI units) under a 5 N force. Instantaneous power is:

- (1) 7 W
- (2) 6 W
- (3) 10 W
- (4) 5 W

Q23. The output Y of the given combinational logic (NAND fed from A and OR-ed with B, finally NOR) is similar to:

- (1) OR gate
- (2) AND gate
- (3) NAND gate
- (4) NOR gate

Q24. A planet's mass is $1/10$ th Earth's and its diameter is half that of Earth. Acceleration due to gravity on it is:

- (1) 4.9 m/s^2
- (2) 3.92 m/s^2
- (3) 19.6 m/s^2
- (4) 9.8 m/s^2

Q25. Statement I: Atoms are electrically neutral as they contain equal +ve and –ve charges. Statement II: Atoms of each element are stable and emit characteristic spectrum.

- (1) I correct, II incorrect
- (2) I incorrect, II correct
- (3) Both correct
- (4) Both incorrect

Q26. A wheel rolls without slipping with linear speed v . P (top) and Q (bottom) of the wheel:

- (1) Both move with equal speed
- (2) Point P has zero speed
- (3) P slower than Q
- (4) P faster than Q

Q27. A particle moving with uniform speed in a circular path maintains:

- (1) Constant velocity but varying acceleration
- (2) Varying velocity and varying acceleration
- (3) Constant velocity
- (4) Constant acceleration

Q28. A thin flat circular disc of radius 4.5 cm placed on water ($S = 0.07 \text{ N/m}$). Excess force to lift it off the surface is:

- (1) 1.98 mN
- (2) 99 N
- (3) 19.8 mN
- (4) 198 N

Q29. In $B = 0.049 \text{ T}$, a magnetic needle does 20 oscillations in 5 s. $I = 9.8 \times 10^{-4} \text{ kg}\cdot\text{m}^2$. If magnetic moment $= x \times 10^{-4} \text{ A}\cdot\text{m}^2$, then x is:

- (1) $50\pi^2$
- (2) $1280\pi^2$
- (3) $5\pi^2$
- (4) $128\pi^2$

Q30. Two equal masses A (velocity v) and B (rest) undergo perfectly inelastic 1-D collision. $v : v_{\text{system}}$ (system velocity) is:

- (1) 4 : 1
- (2) 1 : 4
- (3) 1 : 2
- (4) 2 : 1

Q31. $x = 5 \sin(\pi t + \pi/3) \text{ m}$ represents SHM. Amplitude and time period:

- (1) 5 cm, 1 s
- (2) 5 m, 1 s
- (3) 5 cm, 2 s
- (4) 5 m, 2 s

Q32. Quantities having the same dimensions as solid angle:

- (1) Strain and arc
- (2) Angular speed and stress
- (3) Strain and angle
- (4) Stress and angle

Q33. A thin spherical shell ($R=3 \text{ cm}$, $q=1 \text{ }\mu\text{C}$) is charged. Potential difference between point C (inside) and external point P is:

- (1) $0.5 \times 10^4 \text{ V}$
- (2) zero
- (3) $3 \times 10^4 \text{ V}$
- (4) $1 \times 10^4 \text{ V}$

Q34. A bob is whirled in a horizontal plane at $\omega \text{ rpm}$ with tension T . If speed becomes 2ω with same radius, the new tension is:

- (1) $T/4$
- (2) $\sqrt{2} T$
- (3) T
- (4) $4T$

Q35. A wire of resistance $100 \text{ }\Omega$ is cut into 10 equal parts. First 5 in series, next 5 in parallel; the two combinations in series. The net resistance is:

- (1) $55 \text{ }\Omega$
- (2) $60 \text{ }\Omega$
- (3) $26 \text{ }\Omega$
- (4) $52 \text{ }\Omega$

Section B — Questions 36–50 (Attempt any 10 of 15)

Q36. T-V curves of an ideal gas at three pressures P_1 , P_2 , P_3 compared to Charles's law (dotted) lines. The order is:

- (1) $P_1 > P_2 > P_3$
- (2) $P_2 > P_1 > P_3$
- (3) $P_3 > P_2 > P_1$
- (4) $P_3 > P_1 > P_2$

Q37. Parallel-plate capacitor charged via a resistor (current I). In the gap between plates:

- (1) Displacement current = I , opposite to I
- (2) Displacement current $> I$, any direction
- (3) No current
- (4) Displacement current = I , in same direction as I

Q38. Which is NOT a property of an EM wave in free space?

- (1) Speed = $1/\sqrt{\mu_0 \epsilon_0}$
- (2) Originate from charges moving with uniform speed
- (3) Transverse in nature
- (4) Electric and magnetic energy densities are equal

Q39. Choose the circuit that achieves Wheatstone bridge balance ($10/15 = 10/5$ must hold):

- (1) Option 1
- (2) Option 2
- (3) Option 3
- (4) Option 4

Q40. If plates of a capacitor connected to a battery are moved closer: A. charge increases B. energy decreases C. capacitance increases D. Q/V remains same E. QV increases.

- (1) B, D and E only
- (2) A, B and C only
- (3) A, B and E only
- (4) A, C and E only

Q41. $F = \alpha t^2 + \beta t$ acts on a particle. Which combination is dimensionless?

- (1) $\alpha\beta t$
- (2) $\alpha\beta/t$
- (3) $\beta t/\alpha$
- (4) $\alpha t/\beta$

Q42. Metallic bar ($Y = 0.5 \times 10^{11} \text{ N/m}^2$, $\alpha = 10^{-5} / ^\circ\text{C}$, $L = 1 \text{ m}$, $A = 10^{-3} \text{ m}^2$) heated $0 \rightarrow 100^\circ\text{C}$ without expansion. Compressive force is:

- (1) $100 \times 10^3 \text{ N}$
- (2) $2 \times 10^3 \text{ N}$
- (3) $52 \times 10^3 \text{ N}$
- (4) $50 \times 10^3 \text{ N}$

Q43. Telescope: $f_o = 140 \text{ cm}$, $f_e = 5 \text{ cm}$. Magnifying power for a distant object:

- (1) 17
- (2) 32
- (3) 34
- (4) 28

Q44. An iron bar of length L (moment M) bent at midpoint with arms at 60° to each other. New magnetic moment is:

- (1) $2M$
- (2) $M/\sqrt{3}$
- (3) M
- (4) $M/2$

Q45. A $10\ \mu\text{F}$ capacitor is connected to a $210\ \text{V}$, $50\ \text{Hz}$ source. Peak current is nearly:

- (1) $1.20\ \text{A}$
- (2) $0.35\ \text{A}$
- (3) $0.58\ \text{A}$
- (4) $0.93\ \text{A}$

Q46. Heaters A ($1\ \text{kW}$) and B ($2\ \text{kW}$) connected first in series then in parallel to the same source. Ratio of power outputs (series : parallel):

- (1) $1 : 2$
- (2) $2 : 3$
- (3) $1 : 1$
- (4) $2 : 9$

Q47. v - t plot rises, plateaus, then falls. The corresponding a - t graph is:

- (1) + pulse, then – pulse with plateaus
- (2) + ramp, then – ramp
- (3) Trapezoidal a
- (4) Continuous slow rise

Q48. Pendulum bob mass tripled and length halved. New period = $(x/2) \times$ original. Then x is:

- (1) $2\sqrt{3}$
- (2) 4
- (3) $\sqrt{3}$
- (4) $\sqrt{2}$

Q49. Minimum energy to launch satellite (mass m) from Earth surface (mass M , radius R) into circular orbit at altitude $2R$:

- (1) $GMm/(2R)$
- (2) $GMm/(3R)$
- (3) $5GMm/(6R)$
- (4) $2GMm/(3R)$

Q50. A sheet placed in front of a strong magnetic pole. Force is needed to: A. hold it if magnetic B. hold it if non-magnetic C. move it away with uniform velocity if conducting D. move it away with uniform velocity if non-conducting and non-polar.

- (1) A, C and D only
- (2) C only
- (3) B and D only
- (4) A and C only

CHEMISTRY

Section A — Questions 51–85 (All compulsory)

Q51. Match conversion with Faraday required: A. $1 \text{ mol H}_2\text{O} \rightarrow \text{O}_2$ B. $1 \text{ mol MnO}_2 \rightarrow \text{Mn}^{2+}$ C. 1.5 mol Ca from molten CaCl_2 D. $1 \text{ mol FeO} \rightarrow \text{Fe}_3\text{O}_4 \leftrightarrow \text{I. } 3\text{F II. } 2\text{F III. } 1\text{F IV. } 5\text{F}$

- (1) A-II, B-III, C-I, D-IV
- (2) A-III, B-IV, C-II, D-I
- (3) A-II, B-IV, C-I, D-III
- (4) A-III, B-IV, C-I, D-II

Q52. Which is NOT a redox reaction?

- (1) $\text{H}_2 + \text{Cl}_2 \rightarrow 2 \text{HCl}$
- (2) $\text{BaCl}_2 + \text{Na}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2 \text{NaCl}$
- (3) $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$
- (4) $2 \text{KClO}_3 + \text{I}_2 \rightarrow 2 \text{KIO}_3 + \text{Cl}_2$

Q53. Intramolecular hydrogen bonding is present in:

- (1) 3-nitrophenol (m-)
- (2) HF
- (3) o-nitrophenol
- (4) p-nitrophenol

Q54. Fehling's solution 'A' is:

- (1) Alkaline solution of Rochelle's salt
- (2) Aqueous sodium citrate
- (3) Aqueous copper sulphate
- (4) Alkaline copper sulphate

Q55. $1 \text{ g NaOH} + 25 \text{ mL of } 0.75 \text{ M HCl}$. Mass of NaOH unreacted is:

- (1) Zero mg
- (2) 200 mg
- (3) 750 mg
- (4) 250 mg

Q56. Match shape: A. NH_3 B. BrF_3 C. XeF_4 D. $\text{SF}_6 \leftrightarrow \text{I. Trigonal pyramidal II. Square planar III. Octahedral IV. Square pyramidal}$

- (1) A-III, B-IV, C-I, D-II
- (2) A-II, B-III, C-IV, D-I
- (3) A-I, B-IV, C-II, D-III
- (4) A-II, B-IV, C-III, D-I

Q57. $E^\circ(\text{Mn}^{3+}/\text{Mn}^{2+})$ is more positive than $E^\circ(\text{Cr}^{3+}/\text{Cr}^{2+})$ or $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+})$ because of change of:

- (1) d^5 to d^4
- (2) d^3 to d^4
- (3) d^5 to d^6
- (4) d^5 to d^2

Q58. Match: A. Isothermal B. Isochoric C. Isobaric D. Adiabatic $\leftrightarrow$ I. No heat exchange II. Constant T III. Constant V IV. Constant P

- (1) A-I, B-II, C-III, D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-III, B-IV, C-II, D-I
- (4) A-IV, B-II, C-III, D-I

Q59. Activation energy of a reaction can be calculated knowing:

- (1) Orientation of reactant molecules
- (2) Rate constant at two different temperatures
- (3) Rate constant at standard temperature
- (4) Probability of collision

Q60. C_6H_{14} with two tertiary carbons. IUPAC name:

- (1) 2,3-dimethylbutane
- (2) 2,2-dimethylbutane
- (3) n-hexane
- (4) 2-methylpentane

Q61. Which ions have the same 'spin only' magnetic moment? A. Ti^{3+} B. Cr^{2+} C. Mn^{2+} D. Fe^{2+} E. Sc^{3+}

- (1) B and C only
- (2) A and D only
- (3) B and D only
- (4) A and E only

Q62. Increasing electronegativity of N, O, F, C, Si:

- (1) $O < F < N < C < Si$
- (2) $F < O < N < C < Si$
- (3) $Si < C < N < O < F$
- (4) $Si < C < O < N < F$

Q63. Which alcohol reacts instantaneously with Lucas reagent?

- (1) $(CH_3)_2CH-CH_2OH$ (isobutanol)
- (2) $(CH_3)_3C-OH$ (tert-butanol)
- (3) $CH_3CH_2CH_2CH_2OH$ (n-butanol)
- (4) $CH_3CH_2CH(CH_3)OH$ (sec-butanol)

Q64. Statement I: $[Co(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$ are both octahedral but differ in magnetic behaviour. Statement II: $[Co(NH_3)_6]^{3+}$ is diamagnetic; $[CoF_6]^{3-}$ is paramagnetic.

- (1) I true, II false
- (2) I false, II true
- (3) Both true
- (4) Both false

Q65. I: Boiling order $H_2O > H_2Te > H_2Se > H_2S$. II: Based on M, H_2O should boil low but H-bonding raises BP.

- (1) I true, II false
- (2) I false, II true
- (3) Both true
- (4) Both false

Q66. Match quantum number with information: A. m_l B. m_s C. l D. $n \leftrightarrow$ I. Shape II. Size III. Orientation of orbital IV. Spin orientation

- (1) A-III, B-IV, C-II, D-I
- (2) A-II, B-I, C-IV, D-III
- (3) A-I, B-III, C-II, D-IV
- (4) A-III, B-IV, C-I, D-II

Q67. Match reaction with reagent: A. Bicyclohexyl $\rightarrow$ 2 cyclohexanone B. Benzene $\rightarrow$ benzophenone C. Cyclohexanol $\rightarrow$ cyclohexanone D. Ethylbenzene $\rightarrow$ benzoate $\leftrightarrow$ I. $\text{PhCOCl}/\text{AlCl}_3$ II. CrO_3 III. $\text{KMnO}_4/\text{KOH}/\Delta$ IV. (i) O_3 (ii) $\text{Zn-H}^+/\text{O}$

- (1) A-IV, B-I, C-II, D-III
- (2) A-I, B-IV, C-II, D-III
- (3) A-IV, B-I, C-III, D-II
- (4) A-III, B-I, C-II, D-IV

Q68. Reagent for: cyclohexyl- $\text{CH}_2\text{-CH=CH}_2 \rightarrow$ cyclohexyl- $\text{CH}_2\text{-CH}_2\text{-CHO}$

- (1) (i) BH_3 (ii) $\text{H}_2\text{O}_2/\text{OH}^-$ (iii) alk. KMnO_4 (iv) H^+/O_3
- (2) (i) $\text{H}_2\text{O}_2/\text{H}^+$ (ii) PCC
- (3) (i) $\text{H}_2\text{O}_2/\text{H}^+$ (ii) CrO_3
- (4) (i) BH_3 (ii) $\text{H}_2\text{O}_2/\text{OH}^-$ (iii) PCC

Q69. Reagents that DO NOT react with glucose: A. Tollen's B. Schiff's C. HCN D. NH_2OH E. NaHSO_3

- (1) B and E
- (2) E and D
- (3) B and C
- (4) A and D

Q70. Match molecule with type of C-C bonds: A. Ethane B. Ethene C. C_2 molecule D. Ethyne $\leftrightarrow$ I. $1\sigma + 2\pi$ II. 2π III. 1σ IV. $1\sigma + 1\pi$

- (1) A-III, B-IV, C-II, D-I
- (2) A-III, B-IV, C-I, D-II
- (3) A-I, B-IV, C-II, D-III
- (4) A-IV, B-III, C-II, D-I

Q71. Among Group 16 elements, which does NOT show -2 oxidation state?

- (1) Te
- (2) Po
- (3) O
- (4) Se

Q72. For $2\text{A} \rightleftharpoons \text{B} + \text{C}$, $K_c = 4 \times 10^{-3}$. Currently $[\text{A}] = [\text{B}] = [\text{C}] = 2 \times 10^{-3} \text{ M}$. Then:

- (1) Reaction tends to go backward
- (2) Reaction has gone to completion
- (3) At equilibrium
- (4) Reaction tends to go forward

Q73. Which plot of $\ln k$ vs $1/T$ is consistent with the Arrhenius equation?

- (1) +ve slope line
- (2) Falling line from upper-left
- (3) Rising line with +ve slope through (0, +ve)
- (4) Falling line approaching x-axis

Q74. In which equilibrium are K_p and K_c NOT equal?

- (1) $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$
- (2) $2 \text{BrCl} \rightleftharpoons \text{Br}_2 + \text{Cl}_2$
- (3) $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$
- (4) $\text{H}_2 + \text{I}_2 \rightleftharpoons 2 \text{HI}$

Q75. I: Boiling-point order n-pentane > isopentane > neopentane. II: As branching $\uparrow$, molecule becomes more spherical $\rightarrow$ smaller surface area $\rightarrow$ weaker intermolecular forces $\rightarrow$ lower BP.

- (1) I correct, II incorrect
- (2) I incorrect, II correct
- (3) Both correct
- (4) Both incorrect

Q76. Compound that undergoes SN_1 at the fastest rate:

- (1) Ph-Br (bromobenzene)
- (2) Ph-CH(CH₃)Br (sec)
- (3) Cyclohexyl-CH₂Br
- (4) Cyclohexyl-Br

Q77. Energy of an electron in $n=1$ of He is $-x$ J. Energy of e^- in $n=2$ of Be^{3+} in J is:

- (1) $-4x$
- (2) $-4x/9$
- (3) $-x$
- (4) $-x/9$

Q78. Entropy increases in: A. liquid $\rightarrow$ vapour B. crystal cooled 130 K $\rightarrow$ 0 K C. $2 \text{NaHCO}_3(\text{s}) \rightarrow \text{Na}_2\text{CO}_3(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$ D. $\text{Cl}_2(\text{g}) \rightarrow 2 \text{Cl}(\text{g})$

- (1) A, C and D
- (2) C and D
- (3) A and C
- (4) A, B and D

Q79. Solid $\rightarrow$ vapour without liquid stage. The purification technique based on this principle is:

- (1) Distillation
- (2) Chromatography
- (3) Crystallization
- (4) Sublimation

Q80. Match complex with isomerism type: A. $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)]\text{Cl}$ B. $[\text{Co}(\text{NH}_3)_4(\text{SO}_4)]\text{Br}$ C. $[\text{Co}(\text{NH}_3)_4][\text{Cr}(\text{CN})_6]$ D. $[\text{Co}(\text{H}_2\text{O})_4]\text{Cl}_2 \leftrightarrow$ I. Solvate II. Linkage III. Ionization IV. Coordination

- (1) A-I, B-IV, C-III, D-II
- (2) A-II, B-IV, C-III, D-I
- (3) A-II, B-III, C-IV, D-I
- (4) A-I, B-III, C-IV, D-II

Q81. I: Aniline does not undergo Friedel-Crafts alkylation. II: Aniline cannot be prepared via Gabriel synthesis.

- (1) I correct, II false
- (2) I incorrect, II true
- (3) Both true
- (4) Both false

Q82. Increasing first ionization enthalpy of Li, Be, B, C, N:

- (1) $\text{Li} < \text{Be} < \text{C} < \text{B} < \text{N}$
- (2) $\text{Li} < \text{Be} < \text{N} < \text{B} < \text{C}$
- (3) $\text{Li} < \text{Be} < \text{B} < \text{C} < \text{N}$
- (4) $\text{Li} < \text{B} < \text{Be} < \text{C} < \text{N}$

Q83. Highest number of helium atoms in:

- (1) 4 g of He
- (2) 2.271098 L of He at STP
- (3) 4 mol of He
- (4) 4 u of He

Q84. Most stable carbocation is:

- (1) Cyclopentyl-CH₃
- (2) 1-methylcyclohexyl (3°)
- (3) (CH₃)₂CH-CH₂-CH(CH₃)₂
- (4) (CH₃)C(H)-CH₂-CH(CH₃)CH₃

Q85. K_H of three gases A, B, C in water are 145, 2×10^4 , 35 kbar. Solubility order:

- (1) $A > C > B$
- (2) $A > B > C$
- (3) $B > A > C$
- (4) $B > C > A$

Section B — Questions 86–100 (Attempt any 10 of 15)

Q86. Compound X has 32% A, 20% B, rest C (atomic masses $A=64$, $B=40$, $C=32$). Empirical formula:

- (1) AB_2C_2
- (2) ABC
- (3) A_2BC
- (4) ABC_2

Q87. Products A and B in: $3 \text{ROH} + \text{PCl}_5 \rightarrow 3 \text{RCI} + \text{A}$; $\text{ROH} + \text{PCl}_5 \rightarrow \text{RCI} + \text{HCl} + \text{B}$

- (1) H_3PO_4 and POCl_3
- (2) H_3PO_3 and POCl_3
- (3) POCl_3 and H_3PO_4
- (4) POCl_3 and H_3PO_3

Q88. Plot of osmotic pressure π vs concentration (mol/L) gives slope $25.73 \text{ L}\cdot\text{bar}/\text{mol}$. Temperature ($R = 0.083 \text{ L}\cdot\text{bar}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$):

- (1) 25.73°C
- (2) 12.05°C
- (3) 37°C
- (4) 310°C

Q89. Major product of: PhCH=CHPh (stilbene-like) + $\text{KMnO}_4/\text{H}^+ \rightarrow \text{P}$

- (1) 1,2-diol (PhCH(OH)CH(OH)Ph)
- (2) 1,2-diketone (PhCOCOPh)
- (3) PhCHO
- (4) PhCOOH

Q90. I: $[\text{Co}(\text{NH}_3)_6]^{3+}$ is homoleptic; $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ is heteroleptic. II: $[\text{Co}(\text{NH}_3)_6]^{3+}$ has only one kind of ligand; $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ has more than one.

- (1) I true, II false
- (2) I false, II true
- (3) Both true
- (4) Both false

Q91. Which acid is added during preparation of Mohr's salt to prevent hydrolysis of Fe^{2+} ?

- (1) Dilute HNO_3
- (2) Dilute H_2SO_4
- (3) Dilute HCl
- (4) Concentrated H_2SO_4

Q92. Which is correct?

- (1) $\mu(\text{NF}_3) > \mu(\text{NH}_3)$
- (2) Three canonical forms can be drawn for CO_3^{2-}
- (3) Three resonance structures can be drawn for ozone
- (4) BF_3 has non-zero dipole moment

Q93. Increasing group number ($0 \rightarrow \text{VI}$) for cations: A. Al^{3+} B. Cu^{2+} C. Ba^{2+} D. Co^{2+} E. Mg^{2+}

- (1) E, C, D, B, A
- (2) E, A, B, C, D
- (3) B, A, D, C, E
- (4) B, C, A, D, E

Q94. $\text{CH}_3\text{CH}_2\text{CH}_2\text{I} \rightarrow \text{NaCN} \rightarrow \text{A} \rightarrow \text{OH}^-/\text{partial hydrolysis} \rightarrow \text{B} \rightarrow \text{NaOH}/\text{Br}_2 \rightarrow \text{C (major)}$. C is:

- (1) Butanamide
- (2) α -bromobutanoic acid
- (3) Propylamine
- (4) Butylamine

Q95. Rate quadruples when T changes $27 \rightarrow 57^\circ\text{C}$. E_a ($R = 8.314 \text{ J/K}\cdot\text{mol}$, $\log 4 = 0.6021$):

- (1) 3.80 kJ/mol
- (2) 3804 kJ/mol
- (3) 38.04 kJ/mol
- (4) 380.4 kJ/mol

Q96. $2 \text{NO(g)} \rightleftharpoons \text{N}_2\text{(g)} + \text{O}_2\text{(g)}$ at equilibrium with $[\text{N}_2] = 3.0 \times 10^{-3}$, $[\text{O}_2] = 4.2 \times 10^{-3}$, $[\text{NO}] = 2.8 \times 10^{-3}$. If 0.1 mol/L NO is taken in a closed vessel, degree of dissociation α at equilibrium:

- (1) 0.8889
- (2) 0.717
- (3) 0.00889
- (4) 0.0889

Q97. Reversible isothermal expansion of 1 mol H_2 at 25°C from 20 to 10 atm. Work done ($R = 2.0 \text{ cal/K}\cdot\text{mol}$):

- (1) 413.14 cal
- (2) 100 cal
- (3) 0 cal
- (4) -413.14 cal

Q98. Mass of Cu deposited by 9.6487 A through CuSO_4 for 100 s ($M = 63 \text{ g/mol}$; $1F = 96487 \text{ C}$):

- (1) 31.5 g
- (2) 0.0315 g
- (3) 3.15 g
- (4) 0.315 g

Q99. 2-methylcyclohexanol $\xrightarrow{\text{PBr}_3}$ A (major) $\xrightarrow{\text{alc. KOH}/\Delta}$ B (major). A and B are:

- (1) A: trans-1-Br-2-methyl; B: 2-methyl-cyclohex-1-ene-1-ol
- (2) A: cis-1-Br-2-Me; B: 2-methylcyclohexanone
- (3) A: 1-Br-2-Me-cyclohexane; B: 1-methylcyclohexene
- (4) A: 1-Br-2-Me-cyclohexane; B: 3-methylcyclohexene

Q100. Pair of lanthanoid ions which are diamagnetic:

- (1) Gd^{3+} and Eu^{3+}
- (2) Pm^{3+} and Sm^{3+}
- (3) Ce^{4+} and Yb^{2+}
- (4) Ce^{3+} and Eu^{2+}

BOTANY**Section A — Questions 101–135 (All compulsory)**

Q101. Identify correct statements: A. Vallisneria flowers are colourful and produce nectar B. Waterlily flowers are NOT pollinated by water C. Most water-pollinated species: pollen protected from wetting D. Pollen of some hydrophytes is long, ribbon-like E. In some hydrophytes, pollen is carried passively in water.

- (1) A, C, D and E only
- (2) B, C, D and E only
- (3) C, D and E only
- (4) A, B, C and D only

Q102. Type of conservation where threatened species are taken out of natural habitat and given special care:

- (1) Semi-conservative method
- (2) Sustainable development
- (3) in-situ conservation
- (4) Biodiversity conservation

Q103. Inhibition of succinic dehydrogenase by malonate is a classical example of:

- (1) Competitive inhibition
- (2) Enzyme activation
- (3) Cofactor inhibition
- (4) Feedback inhibition

Q104. Which part of the seed (A=plumule, B=cotyledon, C=radicle, D=...?) is destined to form the root?

- (1) C
- (2) D
- (3) A
- (4) B

Q105. Bulliform cells are responsible for:

- (1) Increased photosynthesis in monocots
- (2) Storing sugars
- (3) Inward curling of leaves in monocots
- (4) Protecting the plant from salt stress

Q106. Required for dark reaction: A. Light B. Chlorophyll C. CO_2 D. ATP E. NADPH

- (1) C, D and E only
- (2) D and E only
- (3) A, B and C only
- (4) B, C and D only

Q107. Formation of interfascicular cambium from fully developed parenchyma cells is an example of:

- (1) Dedifferentiation
- (2) Maturation
- (3) Differentiation
- (4) Redifferentiation

Q108. Hind II cuts DNA at recognition sequence consisting of:

- (1) 4 bp
- (2) 10 bp
- (3) 8 bp
- (4) 6 bp

Q109. Tropical regions show greatest species richness because: A. Tropical latitudes undisturbed for millions of years B. More seasonal C. More solar energy D. Constant environments → niche specialization E. Constant and predictable.

- (1) A, B and E only
- (2) A, B and D only
- (3) A, C, D and E only
- (4) A and B only

Q110. Which is NOT a criterion for fungi classification?

- (1) Mode of spore formation
- (2) Fruiting body
- (3) Morphology of mycelium
- (4) Mode of nutrition

Q111. Number of ATP and NADPH for every CO₂ fixed in the Calvin cycle:

- (1) 3 ATP, 3 NADPH
- (2) 3 ATP, 2 NADPH
- (3) 2 ATP, 3 NADPH
- (4) 2 ATP, 2 NADPH

Q112. Major causes of biodiversity loss: A. Over exploitation B. Co-extinction C. Mutation D. Habitat loss & fragmentation E. Migration

- (1) A, B and E only
- (2) A, B and D only
- (3) A, C and D only
- (4) A, B, C and D only

Q113. Capacity to generate a whole plant from any cell of the plant is:

- (1) Differentiation
- (2) Somatic hybridization
- (3) Totipotency
- (4) Micropropagation

Q114. In Verhulst-Pearl logistic growth $dN/dt = rN[(K-N)/K]$, K indicates:

- (1) Carrying capacity
- (2) Population density
- (3) Intrinsic rate of natural increase
- (4) Biotic potential

Q115. Spindle fibers attach to kinetochores during:

- (1) Anaphase
- (2) Telophase
- (3) Prophase
- (4) Metaphase

Q116. Identify the flower types from figures (a) [ovary above whorls] and (b) [ovary inferior]:

- (1) (a) Perigynous; (b) Epigynous
- (2) (a) Perigynous; (b) Perigynous
- (3) (a) Epigynous; (b) Hypogynous
- (4) (a) Hypogynous; (b) Epigynous

Q117. Match: A. Rhizopus B. Ustilago C. Puccinia D. Agaricus ↔ I. Mushroom II. Smut fungus III. Bread mould IV. Rust fungus

- (1) A-III, B-II, C-I, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-II, C-IV, D-I
- (4) A-I, B-III, C-II, D-IV

Q118. Black seed colour (BB/Bb) is dominant over white (bb). To find the genotype of a black-seed plant, cross it with:

- (1) Bb
- (2) BB/Bb
- (3) BB
- (4) bb

Q119. Pink-flowered Snapdragon × Red-flowered Snapdragon. Phenotypes in progeny:

- (1) Only pink
- (2) Red, pink and white
- (3) Only red
- (4) Red as well as pink

Q120. Match: A. Two or more alternative forms of a gene B. Cross of F1 with homozygous recessive C. Cross of F1 with any parent D. Number of chromosome sets ↔ I. Back cross II. Ploidy III. Allele IV. Test cross

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-I, C-III, D-IV

Q121. Lecithin (small organic compound in living tissues) is an example of:

- (1) Glycerides
- (2) Carbohydrates
- (3) Amino acids
- (4) Phospholipids

Q122. Match: A. Clostridium butylicum B. Saccharomyces cerevisiae C. Trichoderma polysporum D. Streptococcus sp. ↔ I. Ethanol II. Streptokinase III. Butyric acid IV. Cyclosporin-A

- (1) A-III, B-I, C-IV, D-II
- (2) A-IV, B-I, C-III, D-II
- (3) A-III, B-I, C-II, D-IV
- (4) A-II, B-IV, C-III, D-I

Q123. Which component (in the figure) has thin outer walls and highly thickened inner walls (guard cells / vessels)?

- (1) A
- (2) B
- (3) C
- (4) D

Q124. Example of an actinomorphic flower:

- (1) Pisum
- (2) Sesbania
- (3) Datura
- (4) Cassia

Q125. A transcription unit in DNA is defined primarily by three regions (upstream → downstream):

- (1) Inducer, Repressor, Structural gene
- (2) Promoter, Structural gene, Terminator
- (3) Repressor, Operator, Structural gene
- (4) Structural gene, Transposons, Operator

Q126. Fate of a piece of DNA carrying only the gene of interest, transferred into an alien organism: A. Multiplies independently in progeny B. Integrates into recipient genome C. Multiplies and is inherited along with host DNA D. Not an integrated part of chromosome E. Shows ability to replicate.

- (1) B and C only
- (2) A and E only
- (3) A and B only
- (4) D and E only

Q127. Auxin used by gardeners on lawns kills weeds but not grass because auxin:

- (1) Does not affect mature monocots
- (2) Helps cell division in grasses
- (3) Promotes apical dominance
- (4) Promotes leaf abscission

Q128. Cofactor of carboxypeptidase:

- (1) Flavin
- (2) Haem
- (3) Zinc
- (4) Niacin

Q129. Lactose in growth medium is transported into bacteria by action of:

- (1) Permease
- (2) Polymerase
- (3) β -galactosidase
- (4) Acetylase

Q130. Mendel's Law of Dominance explains: A. Of one pair of factors, one is dominant, other recessive B. Alleles do not show expression and both characters appear in F₂ C. Factors occur in pairs in normal diploid plants D. Discrete unit controlling a particular character is called factor E. Only one parental character expressed in monohybrid cross.

- (1) B, C and D only
- (2) A, B, C, D and E
- (3) A, B and C only
- (4) A, C, D and E only

Q131. I: Bt toxins are insect-group specific, coded by gene cry IAc. II: Bt toxin exists as inactive protoxin in *B. thuringiensis*; activated by acidic pH of insect gut.

- (1) I true, II false
- (2) I false, II true
- (3) Both true
- (4) Both false

Q132. I: Parenchyma is living but collenchyma is dead tissue. II: Gymnosperms lack xylem vessels; angiosperms have them.

- (1) I true, II false
- (2) I false, II true
- (3) Both true
- (4) Both false

Q133. I: Chromosomes become gradually visible during leptotene. II: Beginning of diplotene is recognized by dissolution of synaptonemal complex.

- (1) I true, II false
- (2) I false, II true
- (3) Both true
- (4) Both false

Q134. Match: A. Nucleolus B. Centriole C. Leucoplasts D. Golgi ↔ I. Site of glycolipid formation II. Cartwheel organization III. Site of rRNA synthesis IV. Storing nutrients

- (1) A-III, B-IV, C-II, D-I
- (2) A-I, B-II, C-III, D-IV
- (3) A-III, B-II, C-IV, D-I
- (4) A-II, B-III, C-I, D-IV

Q135. List of endangered species was released by:

- (1) FOAM
- (2) IUCN
- (3) GEAC
- (4) WWF

Section B — Questions 136–150 (Attempt any 10 of 15)

Q136. DNA in chloroplast is:

- (1) Linear, single stranded
- (2) Circular, single stranded
- (3) Linear, double stranded
- (4) Circular, double stranded

Q137. What is fused in somatic hybridization (two plant varieties)?

- (1) Protoplasts
- (2) Pollens
- (3) Callus
- (4) Somatic embryos

Q138. Identify (figure shows densely-packed pink spike-like inflorescence with exposed stamens):

- (1) Cleistogamous flowers, autogamy
- (2) Compact inflorescence, complete autogamy
- (3) Wind-pollinated plant inflorescence with well-exposed stamens
- (4) Water-pollinated flowers with mucilaginous covering

Q139. Spraying which PGR on sugarcane increases stem length & yield?

- (1) Cytokinin
- (2) Absciscic acid
- (3) Auxin
- (4) Gibberellin

Q140. Match: A. Frederick Griffith B. Jacob & Monod C. Khorana D. Meselson & Stahl ↔ I. Genetic code II. Semi-conservative replication III. Transformation IV. Lac operon

- (1) A-II, B-III, C-IV, D-I
- (2) A-IV, B-I, C-II, D-III
- (3) A-III, B-II, C-I, D-IV
- (4) A-III, B-IV, C-I, D-II

Q141. Match: A. GLUT-4 B. Insulin C. Trypsin D. Collagen ↔ I. Hormone II. Enzyme III. Intercellular ground substance IV. Glucose transport

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-IV, C-I, D-II
- (3) A-IV, B-I, C-II, D-III
- (4) A-I, B-II, C-III, D-IV

Q142. I: In C₃ plants, some O₂ binds RuBisCO, decreasing CO₂ fixation. II: In C₃ plants, mesophyll cells show very little photorespiration; bundle-sheath cells show none.

- (1) I true, II false
- (2) I false, II true
- (3) Both true
- (4) Both false

Q143. Which step in TCA cycle does NOT involve substrate oxidation?

- (1) Succinyl-CoA → Succinic acid
- (2) Isocitrate → α-KG
- (3) Malic acid → Oxaloacetic acid
- (4) Succinic acid → Malic acid

Q144. Match: A. Citric acid cycle B. Glycolysis C. ETS D. Proton gradient ↔ I. Cytoplasm II. Mitochondrial matrix III. Intermembrane space IV. Inner mitochondrial membrane

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-I, C-IV, D-III

Q145. Which is correct about replication in *E. coli*?

- (1) DDDP polymerizes 5'→3' as well as 3'→5'
- (2) DDDP polymerizes 5'→3'
- (3) DDDP polymerizes only 3'→5'
- (4) DDRP polymerizes only 5'→3'

Q146. If NPP of trophic level 1 = 100x kcal/m²/yr, GPP of trophic level 3 of the same ecosystem is:

- (1) 10x kcal/m²/yr
- (2) (100x/3x) kcal/m²/yr
- (3) (x/10) kcal/m²/yr
- (4) x kcal/m²/yr

Q147. Match: A. Rose B. Pea C. Cotton D. Mango ↔ I. Twisted aestivation II. Perigynous flower III. Drupe IV. Marginal placentation

- (1) A-IV, B-III, C-II, D-I
- (2) A-II, B-III, C-IV, D-I
- (3) A-II, B-IV, C-I, D-III
- (4) A-I, B-II, C-III, D-IV

Q148. Match: A. Robert May B. Alexander von Humboldt C. Paul Ehrlich D. David Tilman ↔ I. Species-area relationship II. Long-term ecosystem expt III. Global species diversity ~7 million IV. Rivet popper hypothesis

- (1) A-I, B-III, C-II, D-IV (or A-III, B-I, C-IV, D-II)
- (2) A-III, B-IV, C-II, D-I
- (3) A-II, B-III, C-I, D-IV
- (4) A-III, B-I, C-IV, D-II

Q149. Match (Stamen type): A. Monoadelphous B. Diadelphous C. Polyadelphous D. Epiphyllous ↔ (Example) I. Citrus II. Pea III. Lily IV. China-rose

- (1) A-I, B-II, C-IV, D-III
- (2) A-III, B-I, C-IV, D-II
- (3) A-IV, B-II, C-I, D-III
- (4) A-IV, B-I, C-II, D-III

Q150. Phaeophyceae correct statements: A. Asexual: biflagellate zoospores B. Sexual: oogamous only C. Stored food: mannitol/laminarin D. Pigments: chl a, c + carotenoids/xanthophyll E. Cellulosic wall + algin coating.

- (1) A, C, D and E only
- (2) A, B, C and E only
- (3) A, B, C and D only
- (4) B, C, D and E only

ZOOLOGY**Section A — Questions 151–185 (All compulsory)**

Q151. Match disease with causal agent: A. Typhoid B. Leishmaniasis C. Ringworm D. Filariasis ↔ I. Fungus II. Nematode III. Protozoa IV. Bacteria

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-IV, C-III, D-I
- (3) A-I, B-III, C-II, D-IV
- (4) A-IV, B-III, C-I, D-II

Q152. Match: A. Non-medicated IUD B. Copper-releasing IUD C. Hormone-releasing IUD D. Implants ↔ I. Multiload 375 II. Progestasert III. Lippes loop IV. LNG-20

- (1) A-IV, B-I, C-II, D-III
- (2) A-III, B-I, C-IV, D-II
- (3) A-III, B-I, C-II, D-IV
- (4) A-I, B-III, C-IV, D-II

Q153. I: Presence/absence of hymen is not a reliable indicator of virginity. II: Hymen is torn during the first coitus only.

- (1) I true, II false
- (2) I false, II true
- (3) Both true
- (4) Both false

Q154. In both sexes of cockroach, anal cerci are present on:

- (1) 8th and 9th segment
- (2) 11th segment
- (3) 5th segment
- (4) 10th segment

Q155. Match: A. Pons B. Hypothalamus C. Medulla D. Cerebellum ↔ I. Additional space for neurons; posture/balance II. Controls respiration & gastric secretions III. Connects brain regions IV. Neurosecretory cells

- (1) A-I, B-III, C-II, D-IV
- (2) A-II, B-I, C-III, D-IV (correct order: A-III, B-IV, C-II, D-I per official key)
- (3) A-II, B-III, C-I, D-IV
- (4) A-III, B-IV, C-II, D-I

Q156. Which is NOT a steroid hormone?

- (1) Progesterone
- (2) Glucagon
- (3) Cortisol
- (4) Testosterone

Q157. Product of DNA-dependent RNA polymerase from template 3' TACATGGCAAATATCCATTCA 5':

- (1) 5' AUGUACCGUUUUAUAGGGAAGU 3'
- (2) 5' ATGTACCGTTTATAGGTAAGT 3'
- (3) 5' AUGUACCGUUUUAUAGGUAAGU 3'
- (4) 5' AUGUAAAGUUUUAUAGGUAAGU 3'

Q158. Match three muscle types (a, b, c) with locations:

- (1) a: Skeletal-Biceps; b: Involuntary-Intestine; c: Smooth-Heart
- (2) a: Involuntary-Nose tip; b: Skeletal-Bone; c: Cardiac-Heart
- (3) a: Smooth-Toes; b: Skeletal-Legs; c: Cardiac-Heart
- (4) a: Skeletal-Triceps; b: Smooth-Stomach; c: Cardiac-Heart

Q159. Stages of cell division: A. G2 B. Cytokinesis C. S D. Karyokinesis E. G1. Correct sequence:

- (1) B, D, E, A, C
- (2) E, C, A, D, B
- (3) C, E, D, A, B
- (4) E, B, D, A, C

Q160. Autoimmune disorders: A. Myasthenia gravis B. Rheumatoid arthritis C. Gout D. Muscular dystrophy E. SLE

- (1) B, C and E only
- (2) C, D and E only
- (3) A, B and D only
- (4) A, B and E only

Q161. Match enzyme with bond hydrolysed: A. Lipase B. Nuclease C. Protease D. Amylase ↔ I. Peptide II. Ester III. Glycosidic IV. Phosphodiester

- (1) A-II, B-IV, C-I, D-III
- (2) A-IV, B-I, C-III, D-II
- (3) A-IV, B-II, C-III, D-I
- (4) A-III, B-II, C-I, D-IV

Q162. Flippers of penguins and dolphins are an example of:

- (1) Convergent evolution
- (2) Divergent evolution
- (3) Adaptive radiation
- (4) Natural selection

Q163. Match respiratory volumes: A. Expiratory capacity B. Functional residual capacity C. Vital capacity D. Inspiratory capacity ↔ I. ERV+TV+IRV II. TV+ERV III. TV+IRV IV. ERV+RV

- (1) A-II, B-I, C-IV, D-III
- (2) A-I, B-III, C-II, D-IV
- (3) A-II, B-IV, C-I, D-III
- (4) A-III, B-II, C-IV, D-I

Q164. Which factor will NOT affect Hardy-Weinberg equilibrium?

- (1) Gene migration
- (2) Constant gene pool
- (3) Genetic recombination
- (4) Genetic drift

Q165. Sequence of human evolution (past → recent): A. Homo habilis B. Homo sapiens C. Homo neanderthalensis D. Homo erectus

- (1) C, B, D, A
- (2) A, D, C, B
- (3) D, A, C, B
- (4) B, A, D, C

Q166. Pathway for action potential through heart: A. AV bundle B. Purkinje fibres C. AV node D. Bundle branches E. SA node

- (1) B, D, E, C, A
- (2) E, A, D, B, C
- (3) E, C, A, D, B
- (4) A, E, C, B, D

Q167. Favourable for oxyhaemoglobin formation in alveoli:

- (1) Low $p\text{CO}_2$ and high H^+
- (2) Low $p\text{CO}_2$ and high temperature
- (3) High $p\text{O}_2$ and high $p\text{CO}_2$
- (4) High $p\text{O}_2$ and lower H^+

Q168. Match: A. α -1 antitrypsin B. Cry IAb C. Cry IAc D. Enzyme replacement therapy $\leftrightarrow$ I. Cotton bollworm II. ADA deficiency III. Emphysema IV. Corn borer

- (1) A-III, B-IV, C-I, D-II
- (2) A-II, B-IV, C-I, D-III
- (3) A-II, B-I, C-IV, D-III
- (4) A-III, B-I, C-II, D-IV

Q169. A: FSH acts on ovarian follicles in female and Leydig cells in male. R: Growing ovarian follicles secrete estrogen; interstitial cells secrete androgen.

- (1) A true, R false
- (2) A false, R true
- (3) Both true; R is correct explanation
- (4) Both true; R is NOT correct explanation

Q170. In pBR322 (E. coli cloning vector with restriction sites EcoRI, ClaI, HindIII, BamHI, Sall, PvuII, PvuI, PstI), genes X and Y respectively are for:

- (1) X: replication; Y: antibiotic resistance
- (2) X: recognition sites; Y: antibiotic resistance
- (3) X: antibiotic resistance; Y: replication
- (4) X: copy-number control; Y: replication

Q171. Match drug with source: A. Cocaine B. Heroin C. Morphine D. Marijuana $\leftrightarrow$ I. Effective sedative in surgery II. Cannabis sativa III. Erythroxylum IV. Papaver somniferum

- (1) A-II, B-I, C-III, D-IV
- (2) A-III, B-IV, C-I, D-II
- (3) A-IV, B-III, C-I, D-II
- (4) A-I, B-III, C-II, D-IV

Q172. Consider: A. Annelids are true coelomates B. Poriferans are pseudocoelomates C. Aschelminthes are acoelomates D. Platyhelminthes are pseudocoelomates

- (1) C only
- (2) D only
- (3) B only
- (4) A only

Q173. I: Descending limb of loop of Henle is impermeable to water and permeable to electrolytes. II: PCT is lined by simple columnar brush-border epithelium for reabsorption.

- (1) I true, II false
- (2) I false, II true
- (3) Both true
- (4) Both false

Q174. Match joints: A. Fibrous B. Cartilaginous C. Hinge D. Ball-and-socket ↔ I. Adjacent vertebrae, limited movement II. Humerus & pectoral girdle, rotational movement III. Skull, no movement IV. Knee, locomotion

- (1) A-II, B-III, C-I, D-IV
- (2) A-III, B-I, C-IV, D-II
- (3) A-IV, B-II, C-III, D-I
- (4) A-I, B-III, C-II, D-IV

Q175. Which is NOT a natural/traditional contraceptive method?

- (1) Lactational amenorrhea
- (2) Vaults
- (3) Coitus interruptus
- (4) Periodic abstinence

Q176. Match: A. Pleurobrachia B. Radula C. Stomochord D. Air bladder ↔ I. Mollusca II. Ctenophora III. Osteichthyes IV. Hemichordata

- (1) A-II, B-IV, C-I, D-III
- (2) A-IV, B-III, C-II, D-I
- (3) A-IV, B-II, C-III, D-I
- (4) A-II, B-I, C-IV, D-III

Q177. Match: A. Axoneme B. Cartwheel pattern C. Crista D. Satellite ↔ I. Centriole II. Cilia and flagella III. Chromosome IV. Mitochondria

- (1) A-II, B-I, C-IV, D-III
- (2) A-II, B-IV, C-I, D-III
- (3) A-IV, B-III, C-II, D-I
- (4) A-IV, B-II, C-III, D-II

Q178. Which is INCORRECT?

- (1) Bio-reactors are used to produce small-scale bacterial cultures
- (2) Bio-reactors have agitator, oxygen delivery and foam control systems
- (3) Bio-reactor provides optimal growth conditions
- (4) Most commonly used bio-reactors are stirring type

Q179. Match prophase-I sub-phase: A. Diakinesis B. Pachytene C. Zygotene D. Leptotene ↔ I. Synaptonemal complex formation II. Completion of terminalisation of chiasmata III. Chromosomes look like thin threads IV. Appearance of recombination nodules

- (1) A-II, B-IV, C-I, D-III
- (2) A-IV, B-III, C-II, D-I
- (3) A-IV, B-II, C-III, D-I
- (4) A-I, B-II, C-IV, D-III

Q180. Match: A. Common cold B. Haemozoin C. Widal test D. Allergy ↔ I. Plasmodium II. Typhoid III. Rhinoviruses IV. Dust mites

- (1) A-III, B-I, C-II, D-IV
- (2) A-IV, B-II, C-III, D-I
- (3) A-II, B-IV, C-III, D-I
- (4) A-I, B-III, C-II, D-IV

Q181. A: Breast-feeding during initial period is recommended for a healthy baby. R: Colostrum contains several antibodies essential for immunity in newborn.

- (1) A correct, R not correct
- (2) A not correct, R correct
- (3) Both correct; R is correct explanation
- (4) Both correct; R is NOT correct explanation

Q182. Match: A. Pterophyllum B. Myxine C. Pristis D. Exocoetus ↔ I. Hag fish II. Saw fish III. Angel fish IV. Flying fish

- (1) A-IV, B-I, C-II, D-III
- (2) A-III, B-I, C-II, D-IV
- (3) A-II, B-I, C-III, D-IV
- (4) A-III, B-II, C-I, D-IV

Q183. 'Ti plasmid' of *Agrobacterium tumefaciens* stands for:

- (1) Tumor inducing plasmid
- (2) Temperature independent plasmid
- (3) Tumour inhibiting plasmid
- (4) Tumor independent plasmid

Q184. Which is NOT a component of the Fallopian tube?

- (1) Infundibulum
- (2) Ampulla
- (3) Uterine fundus
- (4) Isthmus

Q185. Match: A. Down's syndrome B. α -Thalassemia C. β -Thalassemia D. Klinefelter's ↔ I. 11th chromosome II. X chromosome III. 21st chromosome IV. 16th chromosome

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-I, C-II, D-III
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-III, C-IV, D-I

Section B — Questions 186–200 (Attempt any 10 of 15)

Q186. Statements about non-chordates: A. Pharynx perforated by gill slits B. Notochord absent C. CNS is dorsal D. Heart is dorsal if present E. Post-anal tail absent.

- (1) B, D and E only
- (2) B, C and D only
- (3) A and C only
- (4) A, B and D only

Q187. Match: A. Mesozoic Era B. Proterozoic Era C. Cenozoic Era D. Paleozoic Era ↔ I. Lower invertebrates II. Fish & Amphibia III. Birds & Reptiles IV. Mammals

- (1) A-I, B-II, C-IV, D-III
- (2) A-III, B-I, C-IV, D-II
- (3) A-II, B-I, C-III, D-IV
- (4) A-III, B-I, C-II, D-IV

Q188. I: Cerebral hemispheres are connected by corpus callosum. II: The brain stem consists of medulla, pons and cerebrum.

- (1) I correct, II incorrect
- (2) I incorrect, II correct
- (3) Both correct
- (4) Both incorrect

Q189. Spermatogenesis pathway — GnRH → LH and (A) → (B) and (C) → Androgens and Factors → (D = formation of spermatids):

- (1) FSH, Sertoli cells, Leydig cells, spermatogenesis (note: official key gives this as the answer; correct biology has FSH→Sertoli, LH→Leydig)
- (2) ICSH, Leydig, Sertoli, spermatogenesis
- (3) FSH, Leydig, Sertoli, spermiogenesis
- (4) ICSH, Interstitial, Leydig, spermiogenesis

Q190. Match: A. RNA polymerase III B. Termination of transcription C. Splicing of exons D. TATA box ↔ I. snRNPs II. Promoter III. Rho factor IV. snRNAs, tRNA

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-III, C-I, D-II
- (3) A-II, B-IV, C-I, D-III
- (4) A-III, B-II, C-IV, D-I

Q191. Match: A. Exophthalmic goitre B. Acromegaly C. Cushing's syndrome D. Cretinism ↔ I. Excess cortisol, moon face, hyperglycemia II. Hypo-thyroid + stunted growth III. Hyper-thyroid + protruding eye balls IV. Excess GH

- (1) A-III, B-IV, C-II, D-I
- (2) A-III, B-IV, C-I, D-II
- (3) A-I, B-III, C-II, D-IV
- (4) A-IV, B-II, C-I, D-III

Q192. Match: A. Unicellular glandular epithelium B. Compound epithelium C. Multicellular glandular epithelium D. Endocrine glandular epithelium ↔ I. Salivary glands II. Pancreas III. Goblet cells of alimentary canal IV. Moist surface of buccal cavity

- (1) A-III, B-IV, C-I, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-II, B-I, C-III, D-IV
- (4) A-IV, B-III, C-I, D-II

Q193. I: Bone marrow is the main lymphoid organ where all blood cells incl. lymphocytes are produced. II: Both bone marrow and thymus provide micro-environments for development & maturation of T-lymphocytes.

- (1) I correct, II incorrect
- (2) I incorrect, II correct
- (3) Both correct
- (4) Both incorrect

Q194. Match in cockroach: A. Stores food B. Ring of 6-8 blind tubules at foregut-midgut junction C. Ring of 100-150 yellow filaments at midgut-hindgut junction D. Grinds food ↔ I. Gizzard II. Gastric caeca III. Malpighian tubules IV. Crop

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-IV, B-II, C-III, D-I
- (4) A-I, B-II, C-III, D-IV

Q195. Choose the correct statement about juxtamedullary nephron:

- (1) Loop of Henle runs deep into medulla
- (2) Outnumber cortical nephrons
- (3) Located in columns of Bertini
- (4) Renal corpuscle in outer renal medulla

Q196. Match ECG: A. P wave B. QRS C. T wave D. T-P gap ↔ I. Heart muscles electrically silent II. Depolarisation of ventricles III. Depolarisation of atria IV. Repolarisation of ventricles

- (1) A-II, B-III, C-I, D-IV
- (2) A-IV, B-II, C-I, D-III
- (3) A-I, B-III, C-IV, D-II
- (4) A-III, B-II, C-IV, D-I

Q197. Father B■, mother A■, child O■. Possible genotypes: A. $I^A B / I^A i$ / ii B. $I^A B / I^A i$ / $I^A i$ / ii C. $I^A B / I^A i$ / $I^A i$ / $I^A i$ D. $I^A i / I^A i$ / $I^A i / I^A i$ E. $i / I^A B / I^A i$

- (1) C and B only
- (2) D and E only
- (3) A only
- (4) B only

Q198. I: Gause's competitive exclusion principle: two closely related species competing for different resources cannot coexist indefinitely. II: According to Gause, the inferior is eliminated; this may be true if resources are limiting.

- (1) I true, II false
- (2) I false, II true
- (3) Both true
- (4) Both false

Q199. Catalytic cycle of enzyme — correct sequence: A. Substrate-enzyme complex B. Free enzyme ready C. Release of products D. Chemical bonds broken E. Substrate binds active site

- (1) B, A, C, D, E
- (2) E, D, C, B, A
- (3) E, A, D, C, B
- (4) A, E, B, D, C

Q200. I: Mitochondria and chloroplasts are both double-membrane bound organelles. II: Inner membrane of mitochondria is relatively less permeable than chloroplast.

- (1) I correct, II incorrect
- (2) I incorrect, II correct
- (3) Both correct
- (4) Both incorrect

Official Answer Key — NEET (UG) 2024

Answers are referenced to **Code T5** as released by the conducting authority and cross-mapped to **Code T3**.

Q. No.	Ans	Q. No.	Ans	Q. No.	Ans	Q. No.	Ans
1	(4)	51	(1)	101	(1)	151	(4)
2	(1)	52	(4)	102	(4)	152	(2)
3	(3)	53	(4)	103	(4)	153	(1)
4	(2)	54	(3)	104	(3)	154	(1)
5	(4)	55	(1)	105	(3)	155	(2)
6	(2)	56	(1)	106	(2)	156	(1)
7	(3)	57	(3)	107	(1)	157	(1)
8	(3)	58	(3)	108	(1)	158	(3)
9	(3)	59	(2)	109	(1)	159	(3)
10	(3)	60	(4)	110	(1)	160	(4)
11	(2)	61	(3)	111	(3)	161	(4)
12	(4)	62	(2)	112	(1)	162	(4)
13	(3)	63	(2)	113	(3)	163	(4)
14	(4)	64	(2)	114	(4)	164	(2)
15	(2)	65	(3)	115	(2)	165	(1)
16	(1)	66	(3)	116	(1)	166	(1)
17	(1)	67	(3)	117	(1)	167	(1)
18	(4)	68	(3)	118	(4)	168	(3)
19	(3)	69	(3)	119	(3)	169	(3)
20	(3)	70	(4)	120	(1)	170	(4)
21	(1)	71	(4)	121	(2)	171	(2)
22	(2)	72	(2)	122	(2)	172	(4)
23	(1)	73	(3)	123	(3)	173	(1)
24	(4)	74	(1)	124	(2)	174	(3)
25	(4)	75	(3)	125	(1)	175	(4)
26	(3)	76	(3)	126	(3)	176	(2)
27	(4)	77	(2)	127	(4)	177	(1)
28	(1)	78	(1)	128	(2)	178	(2)
29	(2)	79	(3)	129	(3)	179	(2)
30	(1)	80	(2)	130	(4)	180	(2)
31	(4)	81	(1)	131	(3)	181	(4)
32	(2)	82	(3)	132	(2)	182	(2)
33	(2)	83	(4)	133	(1)	183	(2)
34	(4)	84	(2)	134	(4)	184	(4)
35	(3)	85	(3)	135	(4)	185	(4)
36	(3)	86	(3)	136	(4)	186	(3)

37	(4)	87	(2)	137	(4)	187	(4)
38	(2)	88	(2)	138	(3)	188	(1)
39	(4)	89	(4)	139	(3)	189	(1)
40	(4)	90	(3)	140	(2)	190	(3)
41	(4)	91	(3)	141	(4)	191	(2)
42	(4)	92	(3)	142	(1)	192	(1 or 3)
43	(4)	93	(3)	143	(4)	193	(3)
44	(4)	94	(2)	144	(4)	194	(2)
45	(3)	95	(4)	145	(1)	195	(2)
46	(4)	96	(4)	146	(1)	196	(3)
47	(4)	97	(3)	147	(1)	197	(1)
48	(1)	98	(2)	148	(1 or 4)	198	(1)
49	(2)	99	(4)	149	(3)	199	(2)
50	(4)	100	(3)	150	(3)	200	(3)

Marking Scheme: Correct: +4 | Incorrect: -1 | Unattempted: 0

Note: For Q148 and Q192, both options listed in parentheses were considered correct as per the official key.